

ARTICLE



College students with ADHD and their reasons for becoming teachers despite negative childhood experiences

Inbar Levkovich ^a and Zohar Elyoseph^{a,b}

^aFaculty of Graduate Studies, Oranim College of Education, Kiryat Tivon, Israel; ^bThe School of Psychological sciences, Tel Aviv University, Tel Aviv, Israel

ABSTRACT

The purpose of this study was to examine perspectives on childhood, coping methods and reasons for deciding to major in education among undergraduates with Attention Deficit Hyperactivity Disorder (ADHD). While the experiences of children and adolescents with ADHD have been the focus of much research, empirical evidence is lacking regarding the experiences of education students with this disorder. This research seeks to correct this lacuna and in particular to shed light on what motivates students to return to a setting they experienced as so difficult and complex. We used qualitative-phenomenological research methods to examine 30 education students diagnosed with ADHD. Complex childhood experiences emerged during the course of the research. The study participants noted difficulties in remaining attentive, a high level of distractibility, a heightened need to move around and a lack of focus. These symptoms caused them difficulties at school, harmed their self-image and left them with feelings of frustration, failure and jealousy towards their classmates. The participants raised mainly altruistic considerations in explaining why they chose the teaching profession, alongside a negative view of the education system.

ARTICLE HISTORY

Received 3 January 2020
Accepted 29 May 2020

KEYWORDS

ADHD; education students;
teacher education

Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neurocognitive disorder characterised by decreased attentional functions, hyperactive and impulsive behaviour or a combination of both (American Psychiatric Association, 2013; Fuermaier et al., 2012; Nigg, 2013; Polanczyk, Willcutt, Salum, Kieling, & Rohde, 2014). The disorder is defined as a developmental disability that is no longer attributed to children only. Adults with ADHD report a higher incidence of health morbidity, relationship problems, predisposition to addictions and employment instability (Fayyad et al., 2017; Klassen, Miller, & Fine, 2004). ADHD is highly comorbid with other psychiatric disorders (e.g., depression, anxiety, conduct disorder, ASD) (de la Côte-sainte-catherine, 2015). While a percentage of children with ADHD show remission in adulthood, between 15–30% of childhood cases persist into adulthood (Biederman, Petty, O'Connor, Hyder, & Faraone, 2012; Larsson et al., 2013). Previous research shows that adults with ADHD are less likely to pursue higher education

studies than are typically developing controls (DuPaul, Weyandt, O'Dell, & Varejao, 2009; Kuriyan et al., 2013). Many young adults with ADHD switch jobs frequently and have a higher unemployment rate due to their ADHD-related symptoms. They exhibit less job stability, more absences and poorer performance and productivity in the workplace (Barkley & Murphy, 2010; Shifrin, Proctor, & Prevatt, 2010). These adults may experience behaviours marked by disorganisation, hyperactivity and poor concentration (Adamou et al., 2013; Costa Dias et al., 2013).

Studies of college students point to levels of attention difficulties and hyperactivity ranging from 2.9% to 13%, though only 4.5% to 11% acknowledge a current diagnosis of ADHD during their college studies (DuPaul et al., 2017a; Gluting, Sheslow, & Adams, 2002; Gudmundsdottir, Weyandt, & Ernudottir, 2016; Kwak, Jung, & Kim, 2015). The number of college and university students with ADHD in Israel is estimated to be between 2% and 8% of the student population (Fleming & McMahon, 2012; Parker & Boutelle, 2009; The State Comptroller of Israel, 2013).

Academic studies require organisational and self-management skills, the ability to cope simultaneously with multiple assignments, the ability to pay attention to detail, perseverance, and the ability to meet deadlines (Green & Rabiner, 2012). Students with ADHD have more problems with attention, executive functioning, study skills, and social functioning (Barkley & Murphy, 2011; Emmers, Jansen, Petry, van der Oord, & Baeyens, 2017; Jansen et al., 2017; Weyandt et al., 2013). Their difficulties with attention and perseverance, motor activity regulation, and organising and terminating tasks make the requirements of academic study particularly challenging for them (Gray, Fettes, Woltering, Mawjee, & Tannock, 2016). The changing hours and demands typical of the academic environment necessitate long-term organisational and planning skills (Fleming & McMahon, 2012). As a result, students must be capable of making independent choices and setting priorities (Stamp, Banerjee, & Brown, 2014). Research indicates that students with ADHD demonstrate significant difficulties adjusting to the open framework of independent learning typical of academic studies (Feldt, Graham, & Dew, 2011). Most people with ADHD do not believe they are capable of developing effective learning strategies (DuPaul et al., 2009). When students with ADHD do enrol in higher education, they are more likely to have lower test scores and to drop out without finishing their degree (Barkley, Fischer, Smallish, & Fletcher, 2006; DuPaul et al., 2009; Kuriyan et al., 2013). The present study examines childhood perceptions of school and motivations for majoring in education among college students with ADHD.

The motivation levels of students with ADHD are also lower than those of other students (Gray et al., 2016). Analyses of 174 relevant MRI and fMRI studies across a sample of 24,000 revealed significant differences in neural anatomy and processing among those with ADHD compared to unaffected matched controls (Boon, 2020). Some of these students appear to have internalised the environment's negative reactions to them, developed low self-expectations and have trouble believing in their ability to succeed (Kelly & Ramundo, 2006). In addition, some students feel that others do not understand them and consider them incompetent or incapacitated. Moreover, they feel their requests for concessions are met with suspicion (Honkasilta, Vehkakoski, & Vehmas, 2016).

Several factors have been found to help students with ADHD succeed in school, among them accepting their difficulties, being willing to invest in learning, self-belief and internal self-attribution of success and failure (DuPaul et al., 2017a; Honkasilta et al., 2016).

Parental support and involvement in educational and organisational aspects also helped these students succeed (DuPaul et al., 2017a; Wilmshurst, Peele, & Wilmshurst, 2011). Another factor found to be effective is willingness to receive formal assistance from lecturers and informal assistance from peers (DuPaul et al., 2017a).

Childhood experiences as recalled by college students with ADHD

Many students with ADHD recall feeling frustrated, disappointed, unsuccessful, insulted, ashamed and guilty during their school years (Honkasilta et al., 2016; Ohan, Visser, Strain, & Allen, 2011). Despite expectations of academic success from those around them, they repeatedly failed at school. This discrepancy led them to feel guilt, low self-esteem and a sense of ineptitude (DuPaul, Pinho, Pollack, Gormley, & Laracy, 2017b; Gormley, DuPaul, Weyandt, & Anastopoulos, 2016; Scheithauer & Kelley, 2017). A study evaluating the impact of ADHD on employment among adults with this disorder found that many participants recalled negative childhood experiences stemming from their lack of academic success in school. The findings pointed to a lack of optimism about the future, dissatisfaction with many aspects of life, frequent negative thoughts and lower rates of employment and income relative to adults without ADHD (Biederman & Faraone, 2006).

The Israeli context

In Israel, schools are under the supervision of the Ministry of Education (MoE), which has a Special Education Division responsible for setting policy and supervising children with special needs between the ages of 3 and 21 (Nir, Ben-David, Bogler, Inbar, & Zohar, 2016). In the 1950s and 1960s, the trend was to educate children with special needs in separate institutions, and the number of special education schools grew accordingly (Atamni, 2013).

The Israeli Special Education Law legislated in 1988 determined that students with special needs should be mainstreamed in general classrooms wherever possible (Schanin & Reiter, 2007). At the outset of the 21st century the law was amended such that students who were mainstreamed in regular classes would now be entitled to the same funding as children studying in special education schools. Another amendment to the law in 2018 stated that the parents of special education students would be entitled to determine their child's educational placement: a regular class with hours allocated for individual help, a special education classroom in a regular school or a special education school. The objective of this reform was to enable students with prevalent disorders such as ADHD to remain in regular classrooms with hours allocated for individual help (Reiter, 2015). This transition from segregation to inclusion in special education resembles developments in other countries. In Israel, students with prevalent disorders such as ADHD are allocated around four hours of small-group instruction per week. Yet, while Israel embraces the concept of mainstreaming, the significantly smaller number of hours of individual help allocated to mainstreamed students compared to those in special education classrooms and schools has led parents to prefer these settings over mainstreaming. Nevertheless, most students with ADHD are mainstreamed in regular classrooms (Reiter, 2015; Schanin & Reiter, 2007).

In recent years, higher education in Israel and worldwide has also begun to adopt a more inclusive approach. Institutions of higher education in Israel offer accommodations to

students with special needs in the form of customised examinations, accessibility of buildings and study materials, help in professional centres for students with ADHD and subsidised psychological help (Al-Yagon et al., 2013; Lahav, Ben-Simon, Inbar-Weiss, & Katz, 2018).

The current study

Many students with ADHD earn bachelor's degrees in the field of education (Green & Rabiner, 2012). Yet only a few studies have examined the student experience of these education majors or investigated why they want to return to school, a place where they had experiences that negatively affected their self-images (Manjula, Saraswathi, & Padakannaya, 2009). Studies on special education teachers found that people with ADHD chose this profession out of a need to deal with their personal disabilities and ensuing difficulties. They believe that based on their own experience and deep understanding of such disabilities they are in a better position to help their students cope in school. They believe their disabilities help them connect more strongly with their students with special needs. Furthermore, they derive great satisfaction from their work as teachers and even see it as a means of ameliorating the injustice done to them as children (Bastick, 2000; Stephens & Fish, 2010).

Not many studies have examined college students with ADHD and their decision to major in education (Dipeolu, 2011; Nadeau, 2005). The purpose of this study is to examine perceptions of childhood experiences among education majors with ADHD, to explore what they must deal with as students coping with this disability, and to investigate how their experiences as children with ADHD affected their choice of the teaching profession.

This research is important in that it facilitates expanding our understanding of this important group of future teachers who, in addition to having to cope with the symptoms of ADHD in their work as teachers, also are influenced by their memories from school and home in the context of their disability. The findings of the current study can help them understand these memories and the reasons they chose the teaching profession. These insights can assist in constructing ways to support and train these teachers in the teacher education colleges and the schools that hire them and can be used by the professionals who help them in their work.

Methodology

The present study is based on qualitative research set in a phenomenological paradigm. This approach attempts to obtain an in-depth understanding of the phenomenon under examination by entering the world and experiences of the participants (Brinkmann & Kvale, 2015). The qualitative-phenomenological research approach was chosen to enable college students with ADHD to tell their stories and confer meaning on their experiences (Lindseth & Norberg, 2004). The descriptive, inductive, naturalistic, and interpretive method made it possible to highlight the events as experienced subjectively by the participants while focusing on the commonality of a lived experience within a particular group (Creswell & Poth, 2018).

Data collection

The study participants were undergraduate college students majoring in education in Israel. Thirty students participated in the study, ranging in age from 20 to 37 ($M = 26.2$, $SD = .77$). All had been previously diagnosed with ADHD. Their average age at the time of diagnosis was 12.5 years ($SD = .64$). A third of the participants were in their first year of studies at the college, a third were in their second year, 23.3% were in their third year and 10% were in their fourth and final year of studies (Table 1).

Research procedure

The Institutional Review Board of Oranim College of Education, Israel approved the study. The research participants were selected by means of purposeful sampling, which aims to create a diverse sample reflecting a variety of participants, thus providing a comprehensive picture of the subject being investigated (Patton, 2002). Interviewees were recruited through the interviewers’ direct and indirect social connections. They were initially contacted by telephone. Subsequent meetings were organised at the interviewees’ convenience. At the beginning of the interview, participants were told the purpose of the study and how the interview would be conducted. Each participant signed an informed consent form and provided socio-demographic information. Each interview took place at the participant’s home or at a quiet place at the college and lasted about an hour. All interviews were recorded and fully transcribed.

Research tools

The qualitative data in this study were collected through in-depth semi-structured interviews, which served as a means of learning about and understanding the participants’ experiences (Saldaña, 2015). The interviews were conducted based on an interview guide that provided significant but flexible key areas and facilitated dialogue between the interviewer and the interviewee (Creswell & Poth, 2018). Sample questions from the

Table 1. Demographic information about study participants, N = 30.

Variable	n (%)
Gender	
Female	23 (76.7%)
Male	7 (23.3%)
Age	20–37 ($M = 26.2$, $SD = .77$)
Family status	
Single	23 (76.7%)
Married	6 (20%)
Divorced	1 (3.3%)
Age at the time of ADHD diagnosis	
5–8	5 (16.67%)
9–13	12 (40%)
14–18	13 (43.33%)
Year of studies	
1	10 (33.3%)
2	10 (33.3%)
3	7 (23.3%)
4	3 (10%)

interview guide include: “What does ADHD mean to you?”; “What do you recall about yourself as an elementary school student?”; “Is there a difference in the way the disorder manifests itself in childhood and in adulthood?” (Appendix 1).

Data analysis

Data analysis in qualitative-phenomenological research is based on the assumption that all human experiences share a universal essence and that this essence contains meanings that are key to understanding the phenomenon. Such an understanding can emerge from analysing the experiences of different people in the same context. The interviews were analysed using content analysis to identify and encode key themes and patterns. In the first stage, the researchers read all the interviews, noted the major topics emerging from the interviews and created subcategories. They then searched for connecting topics and grouped the subcategories into larger categories. The last stage involved identifying the major themes representing the main research points (Creswell & Poth, 2018).

Results

Qualitative analysis of the interviews revealed three central themes. The first theme – **“Nobody understands me, not even me”: Childhood and adolescence under the shadow of ADHD** – describes life under the shadow of ADHD, including feelings of being impaired, not understood, unmotivated, frustrated and strange. The second theme – **“I was expected to be a failure”: Recollections of school experiences among education majors** – deals with manifestations of ADHD in the students’ lives at school, such as feelings of distraction, delay and failure, and the impact on their relationships with parents and teachers. The third theme – **“I give them what I lacked”: Students’ choice of the teaching profession** – focuses on the students’ professional choice to become teachers and return to school, the very place where they experienced difficulty and frustration.

Theme 1: “Nobody understands me, not even me”: childhood and adolescence under the shadow of ADHD

The childhood recollections of education majors with ADHD are interwoven with a sense of strangeness, inadequacy and frustration emerging from their cumulative experiences of failure. The students described feeling inferior to their peers and even fantasising that one day their difficulties would disappear and they would be like everyone else, without the disorder. Twenty-eight participants noted that as children they felt unmotivated, helpless, angry and despondent and that their fear of failure caused avoidance behaviours. They described dealing with ADHD as a difficult and confusing experience that left its mark on various aspects of their personal, social and academic lives. Because everyone around them saw them as failures, the students developed low expectations of their own abilities. They believed they could not succeed and some even perceived of themselves as stupid:

"My tremendous difficulties in organizing myself caused me a great deal of frustration, self-criticism, an all-time low self-image. In fact, I usually gave up before even trying . . . I've always wanted to be one of those 'do everything' people who are very busy but still manage to do everything. Those who look like life is easy for them because they are organized and know what they want"(27-year-old female third-year student).

All the participating students of all ages described how their perceptions of their ADHD impaired their social and academic functioning. Peer comparison and its impact on self-image are known to be quite significant for school-aged children. The sense of academic inferiority experienced by these students often permeated their social life, where they felt different and even inferior.

In addition to describing their personal difficulties in dealing with ADHD, they also mentioned what their parents went through. Twenty participants told about how their parents had to deal with the loss of their "perfect" child and the consequent loss of themselves as "perfect" parents. The parents begin by denying that their child is having difficulties and gradually feel more and more helpless and hopeless. The participating students described their parents as struggling to understand their difficulties, expecting their child to be "like everyone else" and seeking tailored solutions and answers. Some parents adopted "ignoring" and "normalizing" strategies or struggled to accept how their children dealt with their difficulties.

"My dad was very insistent that I play football. Taking me to play football was really ridiculous. I couldn't even tie my shoelaces until I was quite old. I didn't tie my laces at football practice. I resorted to all sorts of tricks so they looked like they were tied and then pushed them into my shoe"(27-year-old male fourth-year student).

In the transition to adolescence the students' growing self-awareness made their inner experiences more complex. They described how on the one hand ADHD made their lives difficult, frustrating and discouraging, while on the other hand it allowed them to be unique, creative and resourceful. Their "otherness" sometimes became part of their individual identity and even provided them with unique experiences, while at the same time they never felt like they really belonged.

"In some ways ADHD is both my greatest gift and my greatest curse. It's a gift because it gives me an interesting and different way of looking at the world. My brain works differently. I feel I'm very creative because of it. And it's a curse in that I feel like everything is at a very high volume, like my entire life is adolescence . . . It's like the volume has been turned up to the limit in all directions . . . Everything is at its sharpest, and it's hard to take in . . . without any possibility of lowering it to the desired level" (22-year-old female second-year student).

Theme 2: "I was expected to be a failure": recollections of school experiences among education majors

In describing their experiences in elementary and high school, the participating education majors with ADHD mentioned a wide range of symptoms that caused them to be distracted and to have trouble paying attention. Twenty-six participants noted problems in organisation and in following instructions (bringing required materials, preparing homework) and mentioned becoming distracted, procrastinating, behavioural problems and disobeying teachers. These noted difficulties can be classified into management

functions, high cognitive abilities that include planning, monitoring, self-control and feedback, and the ability to allocate attention resources for effective action.

"I was a horrible student. For me, going to school in the morning was only a recommendation, not a requirement. When I did go to school, I did not go into classes. I never brought the appropriate supplies for class like books or pencils and the other kids laughed at me because I always had to borrow these things ... I spent most of my high school years outside smoking cigarettes" (23-year-old female first-year student).

While the ADHD diagnosis affected the lives of some of the interviewees throughout their childhood, others were not diagnosed until much later. As children, they wondered about their difficulties and sought to understand why they experienced school differently from their friends. They reported that their inability to meet school requirements made them feel abnormal. They experienced short-term memory impairment, had more trouble learning than other students, and felt the school had given up on them. They were troubled mainly by not knowing the source of their difficulties:

"I felt really different. Other kids look at you and compare themselves to you. I would ask myself 'Why? What is different about me? Why am I not like everyone else?' ... Sometimes the teacher would send me to the school counselor, I felt they had given up on me" (31-year-old female second-year student).

The participants' learning experiences were also reflected in how their teachers dealt with their classroom difficulties. Some recalled teachers who were impatient and gave up on them, thus making them stop trying to stay in the classroom and learn. Others recalled one particular teacher who gave them a chance to start fresh. The acceptance, caring and desire for them to succeed on the part of this teacher were the main reasons for their success in school:

"Depends on which teachers. Some teachers noticed that it was hard for me and made me feel they were with me. They let me enter the classroom late or go in and out, gave me responsibility, made concessions when it was possible, and made the classroom atmosphere more pleasant. On the other hand, there were the impatient teachers who did not hide their failure to understand my situation and demanded that I meet the same criteria as everyone else, even though they knew I would fail" (27-year-old female fourth-year student). Theme 3: "I give them what I lacked": Students' choice of the teaching profession

In deciding to study education, the participants made a conscious choice to return to a setting that had been difficult for them. Twenty-seven participants indicated that they wanted to help school students with ADHD feel capable, to offer them a variety of activities, understanding, and empathy, to provide them a sense of competence and success.

"My difficulties as a teenager in school and my frustrations because those around me did not understand me led me to choose the teaching profession. From my personal experience with ADHD I acquired many tools and I want to give these to my students. I want to help them, lend them a helping hand and back them up, as many teachers did for me. I want my students to understand that even with ADHD, they can succeed and accomplish great things in life. The experience and knowledge I gained during my attempts to learn put me in an advantageous position. Today I have confidence, knowledge and tools that I can give to my students – there's no one smarter than a person with experience" (37-year-old male third-year student).

The participants found attending school to be difficult and complex. Today they claim they are returning to school to bring about change, to provide their students with a different experience and to close the gaps they experienced as children. They noted that already during their required student teaching in the field, they were able to help their students integrate and succeed in school, compensate for their difficulties and fight for each student without giving up:

"They gave up on me! That's the right word. I didn't want to learn, and it was okay that I wasn't studying. Like it wasn't something wrong. Like, too bad. I don't give up on any of my students. Nobody in any class. They know. I fight for every one of them" (25-year-old male second-year student).

In addition, the decision to major in education revealed another aspect of their ADHD experience. Indeed, some of the participants understand that despite being a burden, ADHD can also be an advantage in their work. Being a teacher can meet their need to be at the centre of things, be creative, energetic and vibrant. Rather than seeing their ADHD as something that sets them apart, they view it a component that makes them special and forms a part of their positive identity. The education majors with ADHD identify with the difficulties experienced by school students with ADHD. They recall their own difficulties at school and want to make things different for their students. The participants stressed their desire not to give up on students but rather to help them succeed and to provide the support they themselves did not receive. Working in teaching and special education can enable the participants to develop a new perspective on themselves and their personal and professional difficulties. Indeed, these education majors have *begun to recognise their unique abilities in the teaching profession*.

Discussion

The purpose of this study was to examine childhood perceptions of school and motivations to major in education among college students with ADHD. The results of this study indicate that the students recalled feelings of confusion, otherness and frustration during childhood, young adulthood and life in general. Their ADHD found expression in distractibility, procrastination and inability to get organised. In addition, students encountered difficulties dealing with their ADHD and the reactions of those around them. They experienced social rejection, were unmotivated, had behavioural problems and felt they were invisible.

The childhood perceptions examined in the present study validate previous findings that the childhood school memories of students with ADHD are fraught with frustration, disappointment, failure, insult, shame, and guilt (Honkasilta et al., 2016; Ohan et al., 2011). Similar findings emerged in studies focusing on adults with ADHD, who reported negative perceptions of childhood as a result of their lack of educational success (Jarrett, 2016). Childhood learning difficulties are common features of ADHD (Caye et al., 2016; Monette, Bigras, & Guay, 2015), and these symptoms continue into adulthood (Sibley et al., 2017). Although prior research has focused mainly on negative consequences (Kuriyan et al., 2013; Raggi & Chronis, 2006), recent studies have highlighted positive aspects of ADHD symptoms, such as close friendship, creativity, vision, originality, resilience and well-being (Glass, Flory, & Hankin, 2012; Lerner, 2016; Wilmshurst et al., 2011).

The participants' experiences in dealing with ADHD during school focused on academic failure, academic conflict, social rejection and a sense of being misunderstood. These experiences are in line with previous research indicating that ADHD manifestations during school evoke feelings of failure and peer group rejection (Harpin, 2005).

The students' descriptions also reflect their sense of embarrassment and inability to comprehend their ADHD symptoms. These descriptions are consistent with previous studies showing that teachers are often extremely frustrated by the classroom behaviour of children with ADHD, especially class interference, aggression and behavioural problems (Rutherford, DuPaul, & Jitendra, 2008). In addition, some studies suggest that teachers generally feel they have poorer relationships with students with ADHD (Fraire, Longobardi, Prino, Sclavo, & Settanni, 2013; Rogers, Belanger-Lejars, Toste, & Heath, 2015). Problems in attention and behaviour, including poor self-control, can negatively impact the quality of the relationships between children and their teachers (Portilla, Ballard, Adler, Boyce, & Obradović, 2014; Rogers et al., 2015). The low self-esteem repeatedly reported by the participants may be a result of social comparison (Harpin, Mazzone, Raynaud, Kahle, & Hodgkins, 2016; Pawaskar, Fridman, Grebla, & Madhoo, 2020). They often report feeling incompetent and incapable compared to other students, being preoccupied with such comparisons and fantasising about succeeding just like the others.

The participants also noted having problems with the academic demands and behavioural norms at school. At the end of elementary school and the beginning of adolescence, children typically adopt a "conventional" moral perspective that stresses obedience to rules and social norms. Thus, the behavioural and cognitive difficulties of children with ADHD in conforming to social norms may also be accompanied by feelings of guilt and shame and by low self-esteem (Kohlberg, 1958).

The participants also reported conflicts in their relationships with their parents, feelings of mutual dissatisfaction, and a sense that their parents did not understand them and could not provide consistent and attentive parenting. These reports are in line with previous reports of irritability, fatigue, and frustration among parents of children with ADHD (Theule, Wiener, Rogers, & Marton, 2011; Wiener, Biondic, Grimpos, & Herbert, 2016). Parents also reported a decline in their quality of life because they tended to be more depressed and to practice more hostile and inconsistent parenting than parents of children without ADHD (Cussen, Sciberras, Ukoumunne, & Efron, 2012; Wiener et al., 2016). In contrast, another qualitative study of college students' perceptions regarding ADHD during their childhood found that parents and teachers were the ones that provided them learning strategies and emotional support (Bartlett, Rowe, & Shattell, 2010). Yet another study examined educators' and parents' attitudes about the best therapeutic responses to children with ADHD. This study found that while teachers felt the most effective intervention was parent-based, parents believed that school intervention was the most effective (Dryer, Kiernan, & Tyson, 2012).

These findings highlight that important adult figures in the lives of these students tend to underestimate the value of their own intervention, hoping that some other form of intervention will bring about change (Honkasilta et al., 2016). The students interviewed in the present study often recognised that their parents and teachers were having trouble understanding their children's difficulties and sometimes did not think they could help.

The present study also examined the reasons the participating students chose the teaching profession. In the interviews, many mentioned that returning to school as

a teacher would provide an opportunity to ameliorate their own experiences and the experiences of their students with ADHD. To the best of our knowledge, the current study is the first to examine the decision to work in education among a population of students with ADHD.

The main considerations the students voiced in the interviews were altruistic (contribution to society, student advancement). They see the role of the teacher in terms of being able to help children with difficulties. They also see the teaching profession as a means of self-expression and self-realisation through internal components of satisfaction, interest and meaning. In other words, the students recognise the value of education, but also understand its potential for self-realisation and for ameliorating their former negative educational experience. In the interviews, they expressed their need to return to school as teachers to correct the issues they faced in the system and to provide their students the kind of adult figure they lacked in their childhood. In their desire to be portrayed as positive and understanding, they often mentioned adults they met in their childhood. Future studies should examine how the childhood experiences of students with ADHD and their reasons for choosing the teaching profession affect their actual experience and functioning as teachers. Of particular interest would be to examine their relations with their students while they themselves continue to cope with ADHD, as well as their relations with authority figures such as the school principal. Moreover, more research is needed about the particular forms of support and help that teacher training institutions should provide to students with ADHD, beyond the academic help provided today.

The study has several limitations. Despite the advantages of qualitative phenomenological research in facilitating in-depth analysis, one of its disadvantages is that it does not allow the inclusion of wide population groups. All the study participants were undergraduate students majoring in education in Israel who reported being diagnosed with ADHD during their years at school. The researchers did not examine these diagnoses. Moreover, some of the participants may have additional disabilities. In the current study, the participants did not provide information about any past or current pharmacological treatments. Future studies should also examine this issue. In addition, the age range of the students who participated in this study was quite broad. Due to the 2002 change in the Special Education Law the participants did not have the same school experience. The younger participants may have been given more hours of individual instruction and been shown more understanding of their ADHD. Their choice of the teaching profession can be explained, among other things, by their memories of their school experiences. Hence, depending on their age, the change in the law and the responses they were given may have influenced their school experience and consequently their decision to choose the teaching profession.

Recommendations for future research

We recommend that future studies examine education students with ADHD from diverse backgrounds studying at different types of institutions. In addition, research should examine the perspectives of family members, peer groups and educators in order to expand and enhance our understanding of this phenomenon from various points of view. To probe the processes in further depth, we recommend including participants' journals in research on education students with ADHD as well as using mixed-method research integrating quantitative and qualitative methods. Longitudinal research can be used to

examine the work of these participants as new teachers at different time points. Perhaps new teachers with ADHD will be found to have unique difficulties and higher burnout and dropout rates. Future assessment studies should examine the needs and responses of these new teachers at school in order to devise appropriate intervention programmes to provide them support at the outset of their careers.

Practical implications

Based on the students' descriptions and on the literature, it appears that children with ADHD feel that adults, including the educators who worked with them, did not understand the manifestations of their disorder and tended to interpret their behaviour stereotypically. Therefore, programmes for training teachers and school counsellors as well as institutions engaged in teacher training must provide information and tools for dealing with ADHD and its educational, emotional, social and behavioural aspects.

Moreover, higher education institutions in Israel should provide assistance to students with ADHD, both on the academic level (such as time planning, test preparation, learning strategies) and on the emotional level, in order to help them succeed in their academic studies. Field training for such students should address fears and concerns along with hopes and empowerment opportunities in anticipation of their work in the school system.

Professionals such as educational counsellors, school principals and educational psychologists who support pre-service teachers as they interact with students with various difficulties are liable to encounter cases in which these pre-service teachers interpret children's behaviour, or the reasons behind this behaviour, in terms of their own childhood experiences ("I was like that too"). While interpretation of this sort can often be helpful because it creates ties between student and teacher, it should be used with caution, for in many cases such interpretation makes it difficult for the teacher to see other interpretations based on knowledge or professional experience. In addition, respondents' reflections can be helpful to current teachers of students with ADHD potentially across all levels of education.

Conclusions

The objective of this study was to examine perspectives on childhood, coping methods and reasons for deciding to major in education among 30 undergraduates with ADHD. The childhood memories of education majors with ADHD about their school experiences are conflictual and marked by a sense that those around them do not understand their disability. For most, their choice of the teaching profession is tied to these experiences, whether they are striving to rectify their own experience or whether they seek to use their personal experience to help other students with difficulties.

Notes on contributors

Inbar Levkovich, PhD, is a Senior Lecturer in the Faculty of Graduate Studies, Oranim Academic College of Education, Israel. Her research interests include stress, trauma, coping and health.

Zohar Elyoseph Lecturer in the Faculty of Education, Oranim Academic College of Education. PhD student in The School of Psychological Sciences Tel Aviv University. Educational Psychologist-intern.

ORCID

Inbar Levkovich  <http://orcid.org/0000-0003-1582-3889>

References

- Adamou, M., Arif, M., Asherson, P., Aw, T. C., Bolea, B., Coghill, D., ... Pitts, M. (2013). Occupational issues of adults with ADHD. *BMC Psychiatry*, 13(1), 59–66.
- Al-Yagon, M., Cavendish, W., Cornoldi, C., Fawcett, A. J., Grünke, M., Hung, L. Y., ... Margalit, M. (2013). The proposed changes for DSM-5 for SLD and ADHD: International perspectives—Australia, Germany, Greece, India, Israel, Italy, Spain, Taiwan, United Kingdom, and United States. *Journal of Learning Disabilities*, 46(1), 58–72.
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Arlington, VA: Author.
- Atamni, K. (2013). Aspects of special education in the world and in Israel: Legislation, implementation and future possibilities. *Artă Și Educație Artistică*, 14(2), 15–21.
- Barkley, R. A., Fischer, M., Smallish, L., & Fletcher, K. (2006). Young adult outcome of hyperactive children: Adaptive functioning in major life activities. *Journal of the American Academy of Child and Adolescent Psychiatry*, 45(2), 192–202.
- Barkley, R. A., & Murphy, K. R. (2010). Impairment in occupational functioning and adult ADHD: The predictive utility of executive function (EF) ratings versus EF tests. *Archives of Clinical Neuropsychology*, 25(3), 157–173.
- Barkley, R. A., & Murphy, K. R. (2011). The nature of executive function (EF) deficits in daily life activities in adults with ADHD and their relationship to performance on EF tests. *Journal of Psychopathology and Behavioral Assessment*, 33(2), 137–158.
- Bartlett, R., Rowe, T. S., & Shattell, M. M. (2010). Perspectives of college students on their childhood ADHD. *MCN: The American Journal of Maternal/Child Nursing*, 35(4), 226–231.
- Bastick, T. (2000). Why teacher trainees choose the teaching profession: Comparing trainees in metropolitan and developing countries. *International Review of Education*, 46(3), 343–349.
- Biederman, J., & Faraone, S. V. (2006). The effects of attention-deficit/hyperactivity disorder on employment and household income. *Medscape General Medicine*, 8(3), 12.
- Biederman, J., Petty, C. R., O'Connor, K. B., Hyder, L. L., & Faraone, S. V. (2012). Predictors of persistence in girls with attention deficit hyperactivity disorder: Results from an 11-year controlled follow-up study. *Acta psychiatrica Scandinavica*, 125(2), 147–156.
- Boon, H. J. (2020). What do ADHD neuroimaging studies reveal for teachers, teacher educators and inclusive education? *Child Youth Care Forum*. <https://doi.org/10.1007/s10566-019-09542-4>
- Brinkmann, S., & Kvale, S. (2015). *Interviews: Learning the craft of qualitative research interviewing* (Vol. 3). Thousand Oaks, CA: Sage.
- Caye, A., Spadini, A. V., Karam, R. G., Grevet, E. H., Rovaris, D. L., Bau, C. H., ... Kieling, C. (2016). Predictors of persistence of ADHD into adulthood: A systematic review of the literature and meta-analysis. *European Child & Adolescent Psychiatry*, 25(11), 1151–1159.
- Creswell, J. W., & Poth, C. N. (2018). *Qualitative inquiry research methods: Choosing among five approaches* (4th ed.). Los Angeles, London, New Delhi: Sage.
- Cussen, A., Sciberras, E., Ukoumunne, O. C., & Efron, D. (2012). Relationship between symptoms of attention-deficit/hyperactivity disorder and family functioning: A community-based study. *European Journal of Pediatrics*, 171(2), 271–280.
- de la Côte-sainte-catherine, C. (2015). ADHD and comorbid disorders in childhood psychiatric problems, medical problems, learning disorders and developmental coordination. *Clinical Psychiatry*, 1(1), 1–9.
- Dias, T. G. C., Kieling, C., Graeff-Martins, A. S., Moriyama, T. S., Rohde, L. A., & Polanczyk, G. V. (2013). Developments and challenges in the diagnosis and treatment of ADHD. *Brazilian Journal of Psychiatry*, 35(suppl 1), S40–S50.

- Dipeolu, A. O. (2011). College students with ADHD: Prescriptive concepts for best practices in career development. *Journal of Career Development, 38*(5), 408–427.
- Dryer, R., Kiernan, M. J., & Tyson, G. A. (2012). Parental and professional beliefs on the treatment and management of ADHD. *Journal of Attention Disorders, 16*(5), 398–405.
- DuPaul, G. J., Dahlstrom-Hakki, I., Gormley, M. J., Fu, Q., Pinho, T. D., & Banerjee, M. (2017a). College students with ADHD and LD: Effects of support services on academic performance. *Learning Disabilities Research & Practice, 32*(4), 246–256.
- DuPaul, G. J., Pinho, T. D., Pollack, B. L., Gormley, M. J., & Laracy, S. D. (2017b). First-year college students with ADHD and/or LD: Differences in engagement, positive core self-evaluation, school preparation, and college expectations. *Journal of Learning Disabilities, 50*(3), 238–251.
- DuPaul, G. J., Weyandt, L. L., O'Dell, S. M., & Varejao, M. (2009). College students with ADHD: Current status and future directions. *Journal of Attention Disorders, 13*(3), 234–250.
- Emmers, E., Jansen, D., Petry, K., van der Oord, S., & Baeyens, D. (2017). Functioning and participation of students with ADHD in higher education according to the ICF-framework: A systematic literature review. *Journal of Further and Higher Education, 41*(4), 435–447.
- Fayyad, J., Sampson, N. A., Hwang, I., Adamowski, T., Aguilar-Gaxiola, S., Al-Hamzawi, A., . . . Gureje, O. (2017). The descriptive epidemiology of DSM-IV adult ADHD in the world health organization world mental health surveys. *ADHD Attention Deficit and Hyperactivity Disorders, 9*(1), 47–65.
- Feldt, R. C., Graham, M., & Dew, D. (2011). Measuring adjustment to college: Construct validity of the student adaptation to college questionnaire. *Measurement and Evaluation in Counseling and Development, 44*(2), 92–104.
- Fraire, M., Longobardi, C., Prino, L. E., Sclavo, E., & Settanni, M. (2013). Examining the student-teacher relationship scale in the Italian context: A factorial validity study. *Electronic Journal of Research in Educational Psychology, 11*(3), 851–882.
- Fuermaier, A. B., Tucha, L., Koerts, J., Mueller, A. K., Lange, K. W., & Tucha, O. (2012). Measurement of stigmatization towards adults with attention deficit hyperactivity disorder. *PloS One, 7*(12), e51755.
- Glass, K., Flory, K., & Hankin, B. L. (2012). Symptoms of ADHD and close friendships in adolescence. *Journal of Attention Disorders, 16*(5), 406–417.
- Glutting, J., Sheslow, D., & Adams, W. (2002). *CARE College ADHD response evaluation manual*. Wilmington, DE: Wide Range Press.
- Gormley, M. J., DuPaul, G. J., Weyandt, L. L., & Anastopoulos, A. D. (2016). First-year GPA and academic service use among college students with and without ADHD. *Journal of Attention Disorders, 23*(14), 1766–1779.
- Gray, S. A., Fettes, P., Woltering, S., Mawjee, K., & Tannock, R. (2016). Symptom manifestation and impairments in college students with ADHD. *Journal of Learning Disabilities, 49*(6), 616–630.
- Green, A. L., & Rabiner, D. L. (2012). What do we really know about ADHD in college students? *Neurotherapeutics, 9*(3), 559–568.
- Gudmundsdottir, B. G., Weyandt, L., & Ernudottir, G. B. (2016). Prescription stimulant misuse and ADHD symptomatology among college students in Iceland. *Journal of Attention Disorders*. doi:10.1177/1087054716684379
- Harpin, V., Mazzone, L., Raynaud, J. P., Kahle, J., & Hodgkins, P. (2016). Long-term outcomes of ADHD: A systematic review of self-esteem and social function. *Journal of Attention Disorders, 20*(4), 295–305.
- Harpin, V. A. (2005). The effect of ADHD on the life of an individual, their family, and community from preschool to adult life. *Archives of Disease in Childhood, 90*(suppl 1), i2–i7.
- Honkasilta, J., Vehkakoski, T., & Vehmas, S. (2016). 'The teacher almost made me cry': Narrative analysis of teachers' reactive classroom management strategies as reported by students diagnosed with ADHD. *Teaching and Teacher Education, 55*, 100–109.
- Jansen, D., Petry, K., Ceulemans, E., van der Oord, S., Noens, I., & Baeyens, D. (2017). Functioning and participation problems of students with ADHD in higher education: Which reasonable accommodations are effective? *European Journal of Special Needs Education, 32*(1), 35–53.
- Jarrett, M. A. (2016). Attention-deficit/hyperactivity disorder (ADHD) symptoms, anxiety symptoms, and executive functioning in emerging adults. *Psychological Assessment, 28*(2), 245–250. <https://doi.org/10.1037/pas0000190>
- Kelly, K., & Ramundo, P. (2006). *You mean I'm not stupid, lazy or crazy?* New York, NY: Scribner.

- Klassen, A. F., Miller, A., & Fine, S. (2004). Health-related quality of life in children and adolescents who have a diagnosis of attention-deficit/hyperactivity disorder. *Pediatrics*, 114(5), e541–e547.
- Kohlberg, L. (1958). *The development of modes of moral thinking and choice in the years 10 to 16*. USA: University of Chicago.
- Kuriyan, A. B., Pelham, W. E., Molina, B. S. G., Waschbusch, D. A., Gnagy, E. M., Sibley, M. H., ... Kent, K. M. (2013). Young adult educational and vocational outcomes of children diagnosed with ADHD. *Journal of Abnormal Child Psychology*, 41(1), 27–41.
- Kwak, Y. S., Jung, Y. E., & Kim, M. D. (2015). Prevalence and correlates of attention-deficit hyperactivity disorder symptoms in Korean college students. *Neuropsychiatric Disease and Treatment*, 11, 797–802.
- Lahav, O., Ben-Simon, A., Inbar-Weiss, N., & Katz, N. (2018). Weekly calendar planning activity for university students: Comparison of individuals with and without ADHD by gender. *Journal of Attention Disorders*, 22(4), 368–378.
- Larsson, H., Asherson, P., Chang, Z., Ljung, T., Friedrichs, B., Larsson, J. O., & Lichtenstein, P. (2013). Genetic and environmental influences on adult attention deficit hyperactivity disorder symptoms: A large Swedish population-based study of twins. *Psychological Medicine*, 43(1), 197–207.
- Lerner, D. A. (2016). Behavioral disinhibition and nascent venturing: Relevance and initial effects on potential resource providers. *Journal of Business Venturing*, 31(2), 234–252.
- Lindseth, A., & Norberg, A. (2004). A phenomenological hermeneutical method for researching lived experience. *Scandinavian Journal of Caring Sciences*, 18(2), 145–153.
- Manjula, P., Saraswathi, G., & Padakannaya, P. (2009). Self-esteem and adjustment among children with reading and writing difficulties. *Studies on Home and Communication Science*, 3(2), 91–95.
- Monette, S., Bigras, M., & Guay, M. C. (2015). Executive functions in kindergarteners with high levels of disruptive behaviors. *British Journal of Developmental Psychology*, 33(4), 446–463.
- Nadeau, K. G. (2005). Career choices and workplace challenges for individuals with ADHD. *Journal of Clinical Psychology*, 61(5), 549–563.
- Nigg, J. T. (2013). Attention-deficit/hyperactivity disorder and adverse health outcomes. *Clinical Psychology Review*, 33(2), 215–228.
- Nir, A., Ben-David, A., Bogler, R., Inbar, D., & Zohar, A. (2016). School autonomy and 21st century skills in the Israeli educational system. *International Journal of Educational Management*, 30(7), 1231–1246.
- Ohan, J. L., Visser, T. A., Strain, M. C., & Allen, L. (2011). Teachers' and education students' perceptions of and reactions to children with and without the diagnostic label "ADHD". *Journal of School Psychology*, 49(1), 81–105.
- Parker, D. R., & Boutelle, K. (2009). Executive function coaching for college students with learning disabilities and ADHD: A new approach for fostering self-determination. *Learning Disabilities Research & Practice*, 24(4), 204–215.
- Patton, M. (2002). *Qualitative evaluation and research methods*. London: Sage.
- Pawaskar, M., Fridman, M., Grebla, R., & Madhoo, M. (2020). Comparison of quality of life, productivity, functioning and self-esteem in adults diagnosed with ADHD and With Symptomatic ADHD. *Journal of Attention Disorders*, 24(1), 136–144.
- Polanczyk, G. V., Willcutt, E. G., Salum, G. A., Kieling, C., & Rohde, L. A. (2014). ADHD prevalence estimates across three decades: An updated systematic review and meta-regression analysis. *International Journal of Epidemiology*, 43(2), 434–442.
- Portilla, X. A., Ballard, P. J., Adler, N. E., Boyce, W. T., & Obradović, J. (2014). An integrative view of school functioning: Transactions between self-regulation, school engagement, and teacher–child relationship quality. *Child Development*, 85(5), 1915–1931.
- Raggi, V. L., & Chronis, A. M. (2006). Interventions to address the academic impairment of children and adolescents with ADHD. *Clinical Child and Family Psychology Review*, 9(2), 85–111.
- Reiter, S. (2015). Meaningful learning in special education teaching and learning based on the cycle of internalized learning: A review. *Open Journal of Social Sciences*, 3(9), 103–112.
- Rogers, M., Belanger-Lejars, V., Toste, J. R., & Heath, N. L. (2015). Mismatched: ADHD symptomatology and the teacher-student relationship. *Emotional and Behavioural Difficulties*, 20(4), 333–348.
- Rutherford, L. E., DuPaul, G. J., & Jitendra, A. K. (2008). Examining the relationship between treatment outcomes for academic achievement and social skills in school-age children with attention deficit hyperactive disorder. *Psychology in the Schools*, 45(2), 145–157.

- Saldaña, J. (2015). *The Coding Manual for Qualitative Researchers*. Sage.
- Schanin, M., & Reiter, S. (2007). From integration to inclusion: The Tirat Carmel Center for Learning Disabilities as a lever for beneficial integration of children with special needs. *Childhood Education*, 83(6), 347–350.
- Scheithauer, M. C., & Kelley, M. L. (2017). Self-monitoring by college students with ADHD: The impact on academic performance. *Journal of Attention Disorders*, 21(12), 1030–1039.
- Shifrin, J. G., Proctor, B. E., & Prevatt, F. F. (2010). Work performance differences between college students with and without ADHD. *Journal of Attention Disorders*, 13(5), 489–496.
- Sibley, M. H., Swanson, J. M., Arnold, L. E., Hechtman, L. T., Owens, E. B., Stehli, A., . . . Vereen, D. (2017). Defining ADHD symptom persistence in adulthood: Optimizing sensitivity and specificity. *Journal of Child Psychology & Psychiatry*, 58(6), 655–662.
- Stamp, L., Banerjee, M., & Brown, F. C. (2014). Self-advocacy and perceptions of college readiness among students with ADHD. *Journal of Postsecondary Education and Disability*, 27(2), 139–160.
- Stephens, T. L., & Fish, W. W. (2010). Motivational factors toward pursuing a career in special education. *Education*, 130(4), 581–594.
- The State Comptroller of Israel. (2013). *Annual report 63g*. Jerusalem: The State Comptroller of Israel. [Hebrew]
- Theule, J., Wiener, J., Rogers, M. A., & Marton, I. (2011). Predicting parenting stress in families of children with ADHD: Parent and contextual factors. *Journal of Child and Family Studies*, 20(5), 640–647.
- Weyandt, L. L., DuPaul, G. J., Verdi, G., Rossi, J. S., Scentsosky, A. J., Vilardo, B. S., . . . Carson, K. S. (2013). The performance of college students with and without ADHD: Neuropsychological, academic, and psychological functioning. *Journal of Psychological Behavioral Assessment*, 35(4), 421–435.
- Wiener, J., Biondic, D., Grimbos, T., & Herbert, M. (2016). Parenting stress of parents of adolescents with attention-deficit hyperactivity disorder. *Journal of Abnormal Child Psychology*, 44(3), 561–574.
- Wilmschurst, L., Peele, M., & Wilmschurst, L. (2011). Resilience and well-being in college students with and without a diagnosis of ADHD. *Journal of Attention Disorders*, 15(1), 11–17.

Appendix 1. Interview questions posed to college students with ADHD

- What does ADHD mean for you?
- What kind of student were you?
- What school subjects did you like? What subjects/topics did you like less and why?
- To what extent did you feel similar to or different from the other students? How did those around you respond to you?
- How did you cope with the demands of school?
- How did you cope with study tasks imposed at school or at home? How did you respond to different challenges?
- Tell me about what led up to your diagnosis. At what age were you diagnosed?
- Did the way your disability manifests itself change between childhood and adolescence?
- How do you cope with life's demands?
- Do you feel there is anything preventing you from achieving your goals? What? Were you ever stopped from working in any particular field or studying any particular areas?
- In what situations does your ADHD manifest itself in your everyday life?
- Why did you choose the teaching profession?
- What are your advantages as a teacher? What are your difficulties?
- Do you feel your disability affects your work as a teacher?
- Tell me about situations in which you experienced your disability in your work.
- Does your disability influence how you work with children? In what way?
- Have you worked with children with ADHD?
- Tell me about your experiences working with these children.
- Do you feel that working with children with ADHD differs from working with other children?

Copyright of Asia-Pacific Journal of Teacher Education is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.